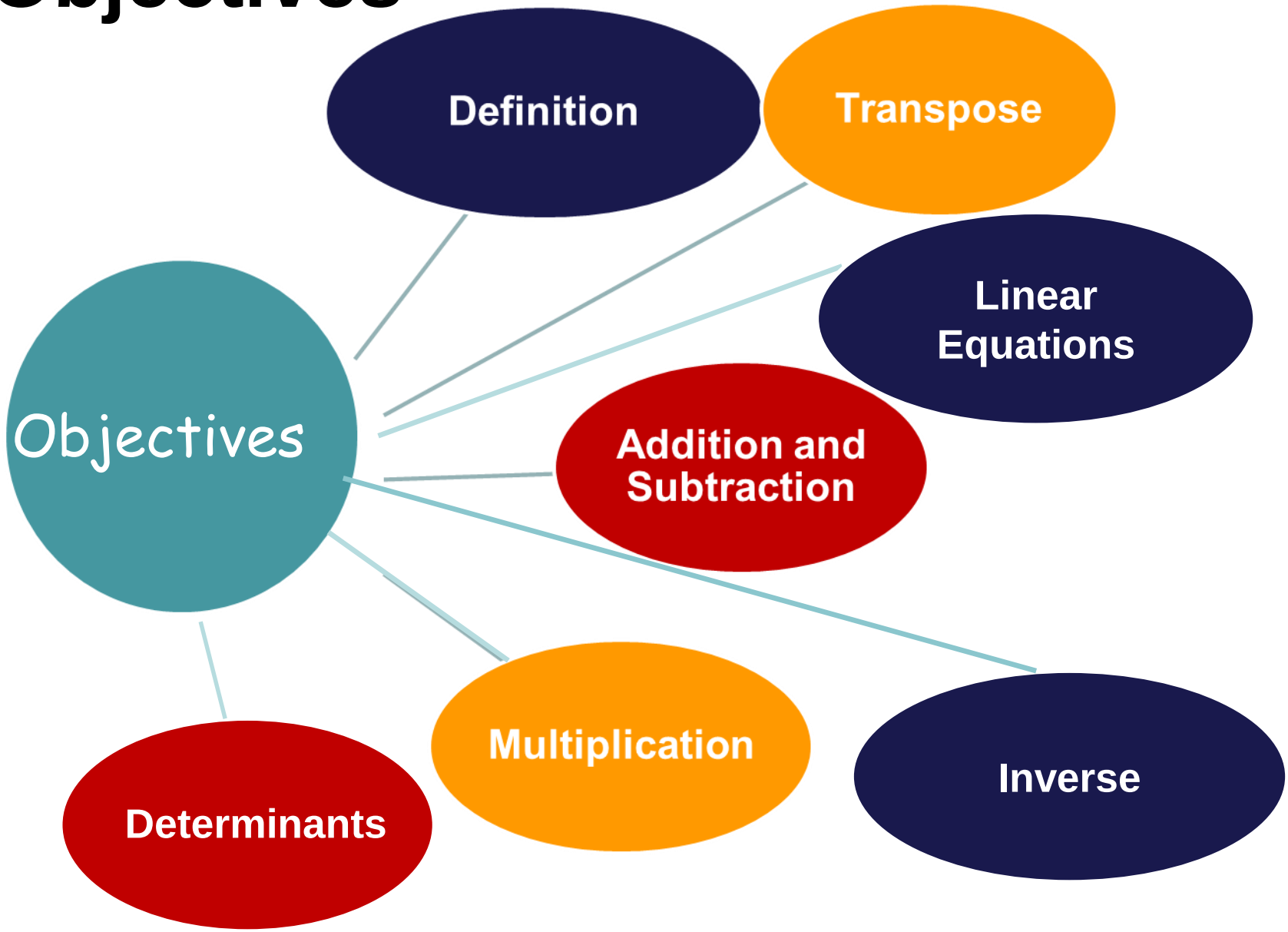
MATRICES

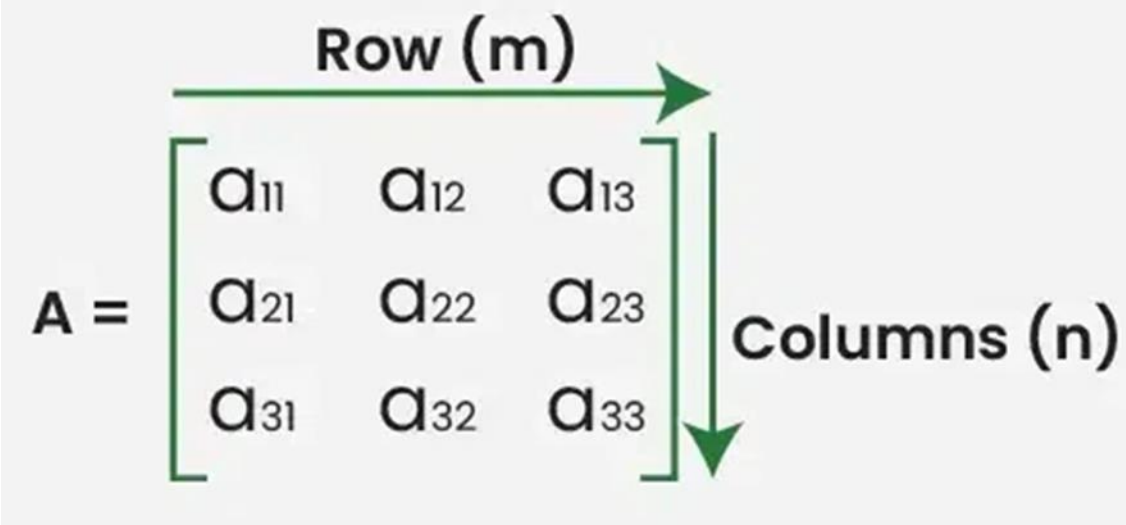
Solve system of equation by using Matrix Algebra



Objectives


Definition
Transpose
Linear Equations
Objectives
Addition and Subtraction
Multiplication
Inverse
Determinants

Definition



Row Matrix

$$\begin{pmatrix} a & b & c \end{pmatrix}$$

Column Matrix
Vector Matrix

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Zero Matrix
Null Matrix

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Diagonal Matrix

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

Scalar Matrix

$$\begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix}$$

Unit Matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Upper Triangular Matrix

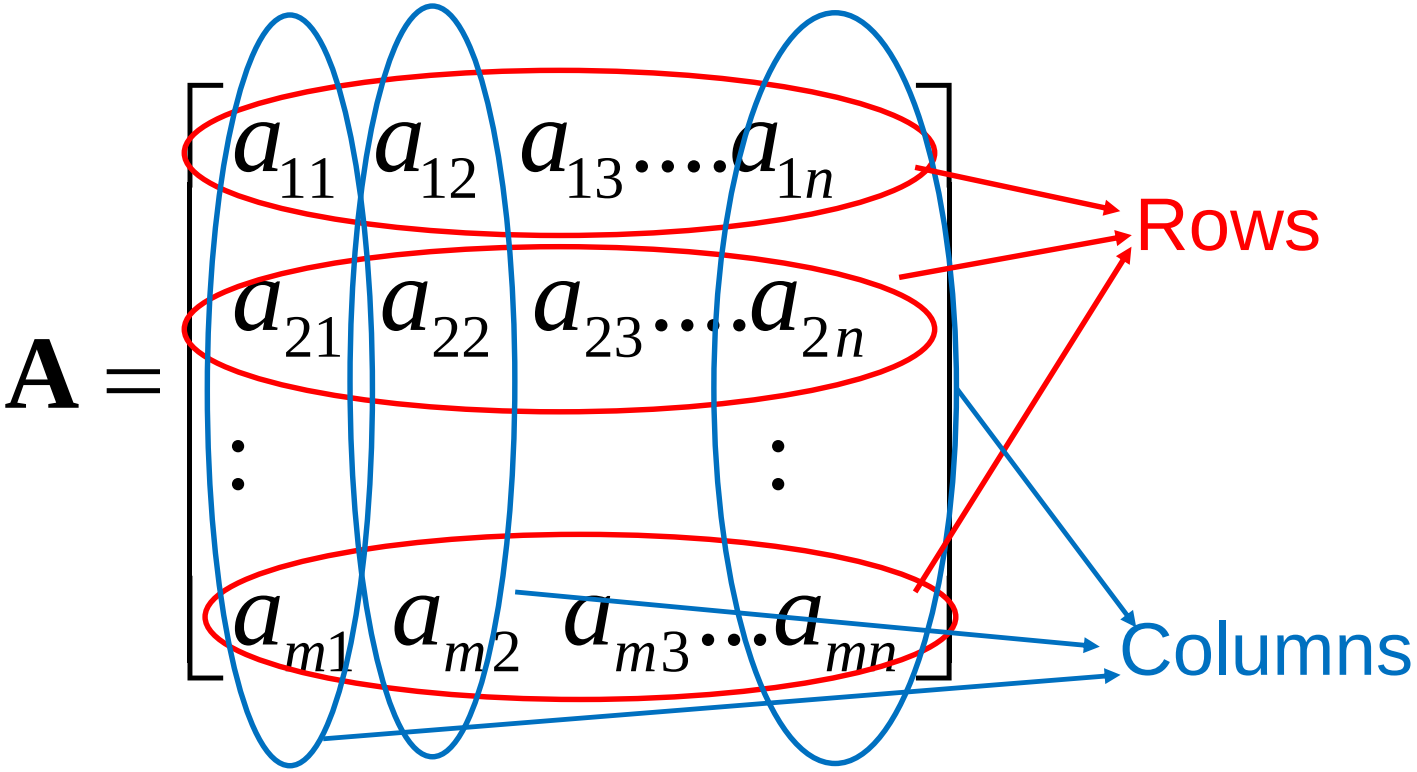
$$\begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$$

Lower Triangular Matrix

$$\begin{pmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{pmatrix}$$

Matrices provide an orderly way of arranging values or functions to enhance the analysis of systems in a systematic manner.

General Form of a Matrix of Size *m* x *n* with *m* Rows and *n* Columns



A matrix is an array of numbers

For example

Three classes are split into boys and girls.

A: 15 boys and 17 girls.

B: 18 boys and 14 girls.

C: 10 boys and 19 girls.

This data could be arranged as a 2 X 3 MATRIX.

$$\begin{pmatrix} 15 & 18 & 10 \\ 17 & 14 & 19 \end{pmatrix}$$

A matrix with R rows and C columns is called an
R X C Matrix.

This is called the ORDER of the matrix.

Each entry in the matrix is called an ELEMENT.

Square Matrix

If *m* = *n*, the matrix is said to be *square.* In this case, the matrix could be designated as an m x m matrix.

$$\mathbf{A_{m,m}}$$

A Matrix with the same number of ROWS as COLUMNS is called a **SQUARE MATRIX.**

$$A = \begin{pmatrix} 5 & 3 \\ 9 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{pmatrix}$$

Vectors

A matrix having only one row is called a *row matrix.*
A matrix having only one column is called a *column matrix.*

A matrix of either form is called a ***vector***. The *row matrix* is called a ***row vector*** and the *column matrix* is called a ***column vector***.

The **CO-ORDINATE** (2,3) has **ONE ROW** and TWO COLUMMS. It is a 1X2 matrix

A matrix with only one row is called a **ROW VECTOR.**

A matrix with only one column and any number of rows is called a **COLUMN VECTOR.**

Types of Matrices

- **Row Matrix** – Has only one row $\begin{bmatrix} 2 & 5 & 7 \end{bmatrix}$
- **Column Matrix** – Has only one column. $\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$
- **Square Matrix** – Same number of rows and columns $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
- **Zero Matrix** – All elements are 0 $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
- **Identity Matrix** – Diagonal elements are 1, others are 0

$1 \times 1 \quad [1]$

$2 \times 2 \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$3 \times 3 \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Example 1. Determine the sizes of the matrixes shown below.

$$\mathbf{A} = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 4 & 6 \end{bmatrix}$$

The matrix has 2 rows and 3 columns. Its size is 2 x 3.

$$\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 7 & -4 \\ 3 & 1 \end{bmatrix}$$

The matrix has 3 rows and 2 columns. Its size is 3 x 2.

Example 2. Determine the size of the matrix shown below.

$$\mathbf{C} = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 6 & 1 \\ -5 & 2 & 1 \end{bmatrix}$$

The matrix has 3 rows and 3 columns. Its size is 3 x 3. It is a ***square*** matrix.

Example 3. Which type of vector is following?

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Addition and Subtraction of Matrices

Matrices can be added together or subtracted from each other ***if and only if they are of the same size***.

Corresponding elements are added or subtracted.

$$\mathbf{C_{m,n} = A_{m,n} \pm B_{m,n}}$$

Matrix Addition

Two matrices of the same size can be added together by adding their corresponding elements.

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

$$B = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 1+5 & 3+7 \\ 2+6 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 10 \\ 8 & 12 \end{pmatrix}$$

Matrix Addition

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

A+B ?

Here, we have $A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$

$$A + B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

$$\Rightarrow A + B = \begin{bmatrix} 1+2 & 2+3 \\ 4+6 & 5+7 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 10 & 12 \end{bmatrix}$$

Matrix Subtraction

Similar to addition, you subtract element-wise

Example: Subtract $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$ **from** $\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$.

Solution:

Let us assume $A = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$ *and* $B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$

$$A - B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$$

$$\Rightarrow A - B = \begin{bmatrix} 2-1 & 3-2 \\ 6-4 & 7-5 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Matrix Subtraction

Similar to addition, you subtract element-wise

$$A = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$$

B-A = ?

Matrix Transpose

The transpose of a matrix is obtained by interchanging rows and columns, respectively.

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \qquad A^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$B = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$$

$B^T = ?$

$$Q = \begin{bmatrix} 1 & 5 \\ 2 & 6 \\ 3 & 8 \end{bmatrix}$$

$$Q^T = \begin{bmatrix} 1 & 5 \\ 2 & 6 \\ 3 & 8 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 8 \end{bmatrix}$$

$$(Q^T)^T = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 8 \end{bmatrix}^T$$

$$(Q^T)^T = \begin{bmatrix} 1 & 5 \\ 2 & 6 \\ 3 & 8 \end{bmatrix} = Q$$

Example 4. Determine the transpose of the matrix **A** below.

$$\mathbf{A} = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 4 & 6 \end{bmatrix}$$

$$\mathbf{A}^{\mathbf{T}} = \begin{bmatrix} 2 & -1 \\ -3 & 4 \\ 5 & 6 \end{bmatrix}$$

If two matrices A and B are of **EQUAL ORDER** then

A + B = B + A

Addition of Matrices (where allowed) is **COMMUTATIVE**

$$\begin{pmatrix} 2 & 1 \\ 3 & -4 \end{pmatrix} + \begin{pmatrix} 6 & -5 \\ 1 & -7 \end{pmatrix} = \begin{pmatrix} 6 & -5 \\ 1 & -7 \end{pmatrix} + \begin{pmatrix} 2 & 1 \\ 3 & -4 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & -4 \\ 4 & -11 \end{pmatrix}$$

EQUAL MATRICES

Two matrices **A** and **B** are equal if **EVERY** element in A is equal to **EVERY** element in B.

Two matrices may be the SAME ORDER but NOT EQUAL.

$$\begin{pmatrix} 4 & 3 \\ -1 & 0 \end{pmatrix} \neq \begin{pmatrix} 4 & 3 \\ 1 & 0 \end{pmatrix}$$

CAN YOU SEE WHY?

Scalars

A **1 x 1** matrix is called a *scalar* and is the form of most variables considered in simple algebraic forms.

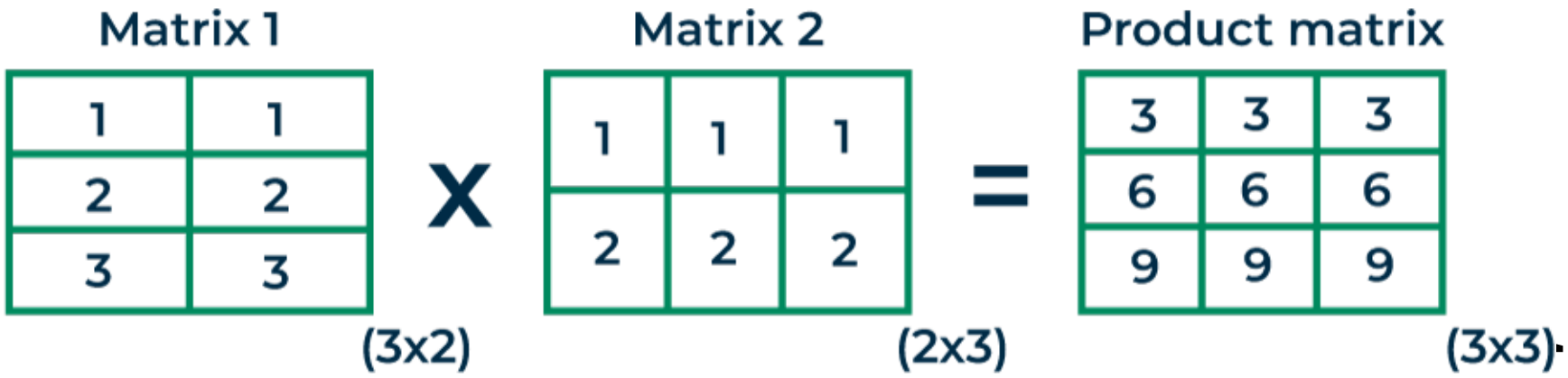
Multiplication by a Scalar

If k is a constant scalar, the matrix k**A** is such that EACH OF ITS ENTRIES ARE MULTIPLIED by k

$$A = \begin{pmatrix} 4 & 3 \\ -1 & 0 \end{pmatrix} \qquad kA = \begin{pmatrix} 4k & 3k \\ -1k & 0k \end{pmatrix} \qquad \tfrac{1}{5}A = \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{-1}{5} & 0 \end{pmatrix}$$

Multiplication of Two Matrices

Two matrices can be multiplied together only if the number of columns of the first matrix is equal to the number of rows of the second matrix.



Multiplication of Two Matrices (Cont.)

The number of rows in the **product matrix** is equal to the number of rows in the first matrix and the number of columns in the product matrix is equal to the number of columns in the second matrix.

$$\mathbf{A}_{\mathbf{m,n}}\mathbf{B}_{\mathbf{n,k}} = \mathbf{C}_{\mathbf{m,k}}$$

Multiplying Matrices

$$\begin{bmatrix} 3 & 4 \\ 7 & 2 \\ 5 & 9 \end{bmatrix} \times \begin{bmatrix} 3 & 1 & 5 \\ 6 & 9 & 7 \end{bmatrix}$$

3×2 ⟵ ⟶ 2×3

Matrix A **Matrix B**

$$\begin{bmatrix} 3 & 2 & 1 & 5 \\ 9 & 1 & 3 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 9 & 0 \\ 1 & 3 & 5 \\ 2 & 4 & 7 \\ 8 & 1 & 5 \end{bmatrix}$$

4 columns **4 rows**

Matrix Multiplication

The product of two square matrices is a new matrix where each element is the sum of the product of elements from the rows of the first matrix and the column of the second matrix.

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

$$B = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$$

$$A \cdot B = \begin{pmatrix} 1(5)+3(6) & 1(7)+3(8) \\ 2(5)+4(6) & 2(7)+4(8) \end{pmatrix} = \begin{pmatrix} 23 & 31 \\ 34 & 46 \end{pmatrix}$$

Division of Matrices ?

There is **no such thing** as division of matrices. However, *matrix inversion* can be viewed in some sense as a procedure similar to division. This process will be considered later.

Example 7. For the following matrices, determine possible orders of multiplication.

$$\mathbf{A} = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 4 & 6 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 7 & -4 \\ 3 & 1 \end{bmatrix}$$

$$\mathbf{AB} = \mathbf{A}_{2,3}\mathbf{B}_{3,2} = \mathbf{C}_{2,2}$$

$$\mathbf{BA} =$$

Example 7. For the following matrices, determine **C=AB**.

$$\mathbf{A} = \mathbf{A}_{2,3} = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 4 & 6 \end{bmatrix} \qquad \mathbf{B} = \mathbf{B}_{3,2} = \begin{bmatrix} 2 & 1 \\ 7 & -4 \\ 3 & 1 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 4 & 6 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 7 & -4 \\ 3 & 1 \end{bmatrix}$$

Example 7. Continuation.

$$c_{11} = (2)(2) + (-3)(7) + (5)(3) = 4 - 21 + 15 = -2$$

$$c_{12} = (2)(1) + (-3)(-4) + (5)(1) = 2 + 12 + 5 = 19$$

$$c_{21} = (-1)(2) + (4)(7) + (6)(3) = -2 + 28 + 18 = 44$$

$$c_{22} = (-1)(1) + (4)(-4) + (6)(1) = -1 - 16 + 6 = -11$$

$$\mathbf{C} = \begin{bmatrix} -2 & 19 \\ 44 & -11 \end{bmatrix}$$

